CIIP Battery Intelligence Report v4.1

India's Green Import Trap

A strategic dependency analysis, industrial resilience framework, and operational intelligence model for electrification-era supply chains.

Author	Framework	Basis	Published
Aditya V Sivaram Poduri	**CIIP Battery Index v4.1**	**Leontief I-O Economics**	**June 2026**

$116B	$19–53B	ZERO	10.0	UBO
Oil Import FY25	**Clean Energy 2030**	**India Li Refining**	**Nickel CIIP Score**	**Core Methodology**
Tier A · IEEFA 2026	Tier C · Scenario range	Tier A · Not a paradox	Tier B · Expert calibrated	Ultimate Beneficiary Ownership

START HERE — THE HUMAN TRANSLATION OF THIS FRAMEWORK

The Story of the Two Cages

Why freeing ourselves from oil may be walking straight into a deeper trap.

THE SITUATION

You live in a house that runs on a rental generator.

Every single day, you buy fuel from a local vendor to keep the lights on. The vendor is difficult, greedy, and changes the price whenever he wants. If you have a bad argument with him, he stops delivering — and your house goes dark that very night. For decades this has been your life.

Then a new company arrives: buy our Solar Power Battery System, they say, and you will never have to buy daily fuel again. The family rejoices. You take out a massive loan, rip out the old wiring, and install the new system. But a few months later, you discover a terrifying truth. You have not actually escaped. You just traded one trap for a much deeper one.

> *Framework connection: The old cage: India's oil dependency — $116B import bill, 15+ countries, fungible commodity. See Section 01.*

PART 1

The Daily Fuel vs. The Fixed Machine

With your old generator, if the vendor shut his shop you could walk down the street and find a different one. The fuel was the same everywhere — fungible.

The new battery system is different. It requires specific, rare spare parts to keep running. If the foreign company gets into a fight with you, they do not just stop selling daily fuel — they block the spare parts. Your expensive system becomes a useless block overnight. With oil, you paid a daily tax. With the battery, you handed the master keys of your entire house to a single stranger.

> *Framework connection: Oil = flow commodity, recoverable in weeks. Battery minerals = capital goods lock-in. Failure mode is CapEx stranding, not petrol queues. See Section 02.*

PART 2

The Illusion of 'Made in My Garage'

To save face, you start building batteries yourself. Assembly line. Local workers. 'Built at Home' on the box. But when you open the components, you realise you are cheating yourself. The core magical powder inside the battery cannot be made in your garage. You have to import that exact chemical powder from a giant monopoly.

All you are doing is putting the imported pieces together, screwing the lid shut, and polishing the box. If the monopoly stops shipping the powder, your grand factory grinds to a halt on day one. Your independence is a cosmetic illusion.

> *Framework connection: The magical powder = $NiSO_4$, $CoSO_4$, $MnSO_4$ metal sulphates. India's pCAM reactors perform the precipitation step. Everything that matters — the metal sulphates — is imported from Chinese-controlled processing. See Section 03.*

PART 3

Someone runs in: 'Good news! We found a massive pile of raw, unmined rock in our backyard that contains the exact ingredients for the magical powder!' Everyone celebrates. But then reality sets in. It is just raw rock buried deep under a mountain in a dangerous, unstable area. You do not have the massive chemical ovens or the secret recipe to turn that rough rock into 99.9% pure battery powder.

To use it, you would have to ship the dirt to the monopoly, pay them to refine it, and buy it back at a massive premium. Finding the raw material does not mean you are rich. It just highlights how helpless you are without the tools to process it.

> *Framework connection: The gold mine = India's J&K; lithium. G3/G4 UNFC inferred resource. Claystone deposit. No proven extraction technology globally. 13-20 years minimum to battery-grade output. Not a paradox. A policy delusion. See Section 05.*

PART 4

You try to be smart. 'I will buy from an independent farmer in Australia and a reputable shop in Europe.' It looks great on paper. Your shipping receipts show different country flags.

But when you look closely at the bank accounts, you find that the European shop and the Australian farm are heavily financed and secretly controlled by the exact same monopoly you were trying to avoid. The name on the delivery truck changed. The hands pulling the strings remained exactly the same.

> *Framework connection: Country-of-origin certificates are customs instruments, not risk instruments. Indonesian HPAL plants owned by Tsingshan and Huayou (Chinese). Belgian $NiSO_4$ feedstock from Russian/African mines with Chinese concessions. The methodology to see through it: Ultimate Beneficiary Ownership (UBO) Risk analysis. See Section 04.*

THE THREE MANDATORY SOLUTIONS

1 **Build the Refining Ovens, Not Just the Assembly Lines**

Stop celebrating factories that only screw battery boxes together. Build the high-tech chemical plants that turn raw ore into pure battery-grade metal sulphates — $NiSO_4$ and $CoSO_4$ sulphation capacity. That is the step India has zero capacity at today.

2 **Set Up the Ultimate Recycling System**

Treat every imported battery like gold. When it dies, hydrometallurgical recycling must extract every milligram of precious material. The metals stay in India's industrial ecosystem permanently — a secondary supply that bypasses the monopoly entirely.

3 **Change the Recipe**

Shift large systems aggressively to LFP and Sodium-ion chemistries for appropriate applications. These bypass nickel and cobalt dependency. Sodium-ion uses sodium — sourceable domestically — completely breaking the foreign chokehold for grid storage and two-wheeler segments.

> ***"We are currently walking out of a cage made of oil and straight into a cage made of technology and rare minerals. This research is the blueprint that cuts the bars of the new cage — before the door locks us in for the next twenty years."***

The analytical proof of every sentence above begins in Section 01. — Aditya V Sivaram Poduri

ES Executive Summary — Five Core Conclusions

This framework answers one question: what second-order industrial dependencies emerge when a civilisation electrifies? The five conclusions below represent the institutional-grade findings.

1 — Dependency Is Migrating, Not Disappearing	India's electrification transition may reduce crude oil dependency while simultaneously increasing exposure to foreign-controlled refining, battery chemistry, and industrial processing ecosystems. The dependency does not disappear. It substitutes.
2 — The Strategic Chokepoint Is Midstream Processing	Mine ownership alone does not determine industrial sovereignty. Refining, sulphation, precursor chemistry, process equipment, and manufacturing specifications are the dominant leverage layers — all concentrated in China.
3 — Industrial Expansion Risk Matters More Than Immediate Consumption Risk	The primary risk is not daily economic collapse. It is long-term industrial expansion paralysis through delayed commissioning, stranded manufacturing CapEx, and processing bottlenecks. A Growth Chokepoint, not a Daily Fuel Trap.
4 — Apparent Diversification May Fail Under UBO Analysis	Country-of-origin diversification does not represent true strategic diversification when production allocation decisions remain controlled by the same upstream ownership ecosystem. UBO Risk analysis is required.
5 — The Core Framework Is About Industrial Architecture	CIIP is not a commodity price forecasting model. It is an industrial dependency intelligence framework designed to identify hidden chokepoints, technological lock-in, and Leontief propagation pathways during electrification.

01 Central Thesis

This framework answers one question: what second-order industrial dependencies emerge when a civilisation electrifies? The answer exposes a structural risk that India's current policy architecture is not designed to detect — because it is invisible at the country-of-origin level and only visible at the molecular processing level.

> ***With oil, you paid a daily tax. With the battery, you handed the master keys of your entire house to a single stranger.***
>
> The Two Cages · Section 00 · The analytical proof follows

> ***"Critical mineral dependency may become more strategically rigid and technologically irreversible than oil dependency — despite lower headline import values. Once India's EV and solar industries are built around Chinese cell chemistry, Chinese wafer specifications, and Chinese process equipment, switching is not a 6-month logistics problem. It is a 10-year industrial rebuild. The supplier changes from Riyadh to Beijing. The leverage does not disappear — it becomes structurally difficult and economically costly to reduce within short industrial timelines."***

WHY OIL AND MINERAL DEPENDENCY ARE STRUCTURALLY DIFFERENT — NOT EQUIVALENT

Oil is a flow commodity consumed continuously — every barrel imported creates no lock-in for tomorrow. Critical minerals go into capital goods with 8–15 year lifecycles. A battery gigafactory built on Chinese LFP technology in 2026 runs on that technology until 2038+. The dependency is structural and technologically embedded, not annual and substitutable. Battery-grade lithium hydroxide processed to Chinese specification has no short-term substitute for factories built around that specification. The correct claim: critical mineral dependency may prove more strategically rigid and harder to exit — not that headline import values are identical.

02 The Growth Chokepoint — Not a Daily Fuel Trap

The most common misreading of this framework: 'India's grid runs today, India's vehicles run today — where is the crisis?' That question betrays a fundamental confusion between two structurally different risk categories.

> ***India's existing economy continues to function. India's next economy — the electrified, domestically manufactured, clean energy economy — cannot be built. That is the crisis this framework is designed to prevent.***
>
> Section 02 · Industrial expansion trap — not daily fuel trap

THE PRECISE FAILURE MODE — CAPEX STRANDING

India's existing vehicle fleet operates today. The critical mineral risk manifests as industrial expansion paralysis. An Indian infrastructure fund commits Rs 8,000 crore to a battery gigafactory in Gujarat. Construction completes in 2028. The plant is ready. The precursor chemical supply contracts — $NiSO_4$ from Belgium, $CoSO_4$ and $MnSO_4$ from Chinese processors, graphite from Chinese suppliers — are either restricted by Chinese export controls, redirected by Chinese corporate ownership decisions at Indonesian HPAL plants, or priced at a level that destroys cell economics against imported Chinese cells. The Rs 8,000 crore is stranded. The capacity exists on paper. It cannot operate at viable economics.

THE TIMELINE ASYMMETRY THAT MAKES THIS UNIQUELY DANGEROUS

When India's crude oil supply is disrupted, strategic petroleum reserves cover 9–14 days. Alternative crude routing takes 3–6 weeks — the harm is fast and recoverable. When India's battery-grade lithium hydroxide supply is disrupted, there is no strategic reserve, no spot market, no alternative refiner at scale outside Chinese processing, and a 4–7 year greenfield commissioning timeline for any new domestic capacity. The asymmetry between disruption speed (immediate) and recovery speed (years to decades) is what makes this structurally more dangerous for long-duration industrial planning.

Recovery timeline comparison: weeks to restore 80% supply after disruption event (Tier B)

03 The pCAM Molecular Chokepoint

When India builds a domestic cathode precursor plant and claims midstream localisation, the analytical question is not where the reactor sits. It is what enters the reactor.

> ***All you are doing is putting the imported pieces together, screwing the lid shut, and polishing the box. If the monopoly stops shipping the powder, your grand factory grinds to a halt on day one.***
>
> The Two Cages · Part 2 · The pCAM optical illusion — molecular proof below

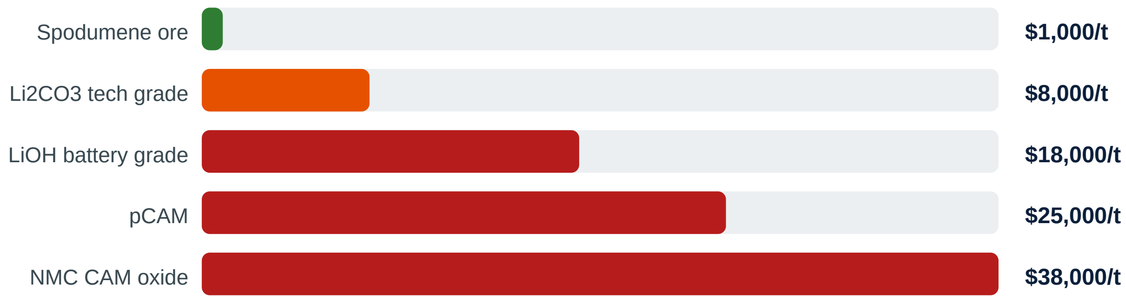
THE PCAM SYNTHESIS — WHAT ACTUALLY HAPPENS IN AN INDIAN REACTOR

Battery-grade NMC cathode active material (CAM) requires two-stage synthesis. Stage one produces the precursor cathode active material (pCAM) by feeding NiSO4-6H2O, CoSO4-7H2O, and MnSO4-H2O dissolved in deionised water into a CSTR reactor. Sodium hydroxide (NaOH) raises pH to 11–12, co-precipitating Ni(OH)2, Co(OH)2, and Mn(OH)2 as a mixed hydroxide precursor (MHP) particle. Ammonium hydroxide (NH4OH) acts as chelating agent — complexing with metal ions to produce uniform spherical secondary particles of 10–15 microns that determine tap density and cell energy density. The NaOH precipitation increases bulk mass and volume, creating the appearance of a larger domestically produced intermediate. The import dependency has been pushed one step upstream, made invisible in trade statistics, and wrapped in a domestic manufacturing claim.

Layer	Step	Process Description	India Capacity
Layer 2a	Smelting & Refining	Ore to Class I Nickel / Li Carbonate	ZERO
Layer 2b	Sulphation / Conversion	Class I Metal to Battery-Grade Metal Sulphates (NiSO4, CoSO4, MnSO4)	ZERO
Layer 2c	pCAM Synthesis	Metal Sulphates + NaOH + NH4OH to MHP — co-precipitation reactor	NASCENT (inputs 100% imported)
Layer 2d	CAM Calcination	pCAM + LiOH to NMC Cathode Oxide (700–900C, O2 atmosphere)	LIMITED (LiOH 74% Chinese)
Layer 3	Cell Assembly	Electrodes + Electrolyte + Separator in Dry Room	NASCENT (equipment 80–90% Chinese)

THE OPTICAL ILLUSION — PRECISELY DIAGNOSED

An Indian pCAM plant is a reaction vessel that converts imported Chinese-processed metal sulphates into a mixed hydroxide powder using domestically available NaOH and NH4OH. The domestic value addition is the reactor operation and precipitation chemistry. The strategic value addition is limited under current upstream dependency conditions — because the critical chemical intelligence, the processing complexity, and the geopolitical leverage all reside in the metal sulphate inputs. India has localised the least strategic part of the midstream chain. The solution is not to build more pCAM reactors. It is to build metal sulphate production capacity — NiSO4 and CoSO4 sulphation plants. India has zero capacity at this step.



Value addition ladder — the export-reimport trap. Spodumene at $1,000/t reimported as LiOH at $18,000/t. 18x value destruction if India mines without refining. (Tier B)

04 Ultimate Beneficiary Ownership Risk (UBO Risk)

The single most important methodological contribution of this framework: geopolitical supply chain risk cannot be assessed by reading the flag on the shipping container. The operationally relevant question is who makes the production allocation decision.

> ***The name on the delivery truck changed. The hands pulling the strings remained exactly the same.***
>
> The Two Cages · Part 4 · The fake map — UBO analysis dissolves it below

UBO RISK — FORMAL DEFINITION

Ultimate Beneficiary Ownership Risk (UBO Risk): The supply chain vulnerability created when the corporate entity that makes production, pricing, and allocation decisions for a critical industrial input is beneficially owned or operationally controlled by a foreign state-linked conglomerate — regardless of the geographic location of the physical asset or the national flag on the shipping certificate. A country-of-origin certificate is a customs instrument designed to assess tariff liability. It was not designed to assess strategic vulnerability. Corporate strategists who rely on country-of-origin certificates as their geopolitical risk tool are measuring the wrong variable.

NICKEL VARIANT A VS VARIANT B — TWO CHEMICAL MARKETS, TWO RISK STRUCTURES

India's nickel exposure operates across two chemically distinct segments. Variant A (Nickel Oxides and Hydroxides / MHP): India sources 64.7% from Australia, remainder from Indonesia. Tier A — IEEFA/ICRIER. Variant B (Nickel Sulphate NiSO4-6H2O battery-grade): India sources 65%+ from Belgium (Umicore), with Japan and Finland as secondary suppliers. Tier A — IEEFA/ICRIER. Conflating these two markets produces the wrong policy prescription.

Supply Node	Origin Country	Beneficial Owner	UBO Nationality	Verdict
Indonesian HPAL Plants	Indonesia	Tsingshan / Huayou Cobalt	CHINA	Indonesian certificate, Chinese control
Belgian NiSO4 (Umicore)	Belgium	Umicore (Belgian public)	MIXED	Corporate: Belgian. Feedstock: Russian/African-Chinese
DRC Cobalt (Tenke/Kisanfu)	DRC	CMOC (Chinese state)	CHINA	Geographic: DRC. Beneficial owner: Chinese
Australia BHP Nickel West	Australia	BHP (Anglo-Australian)	WESTERN	Genuine Western UBO — benchmark for independence

THE DIVERSIFICATION MIRAGE — DISSOLVED AT UBO LEVEL

The Indonesia mirage: HPAL plants are Indonesian assets with Indonesian export certificates. Beneficial owners are Tsingshan and Huayou — Chinese state-influenced conglomerates. Production allocation decision (sell MHP to India or redirect to Chinese cathode plants) is made in China. The Belgium mirage: Umicore is genuinely Belgian — real first-order diversification. But Umicore's NiSO4 feedstock includes Norilsk-linked and African mines with Chinese concession ownership. First-order corporate diversification holds. Second-order feedstock diversification partially fails. India's apparent diversification survives a country-of-origin audit and partially fails a UBO audit.

05 The J&K Lithium Reality Check

Not a Paradox. Not an Irony. A Policy Delusion.

Previous versions of this framework cited India's 5.9 million tonne J&K; lithium occurrence as an 'astounding paradox.' That framing was analytically lazy. This version corrects it with precision.

> ***Finding the raw material does not mean you are rich. It just highlights how helpless you are without the tools to process it.***
>
> The Two Cages · Part 3 · The unreachable gold mine — engineering reality below

THE UNFC CLASSIFICATION — WHAT THE NUMBERS ACTUALLY MEAN

The Geological Survey of India classified the J&K lithium occurrence under UNFC as G3/G4 — reconnaissance-level geological evidence. G3 means the deposit has been identified through surface mapping, geochemical sampling, and limited drilling but has not been systematically delineated through close-spaced infill drilling. G4 is purely speculative occurrence. Neither classification constitutes a proven, measured, or indicated reserve under any internationally recognised mining finance standard — not JORC, not NI 43-101, not SAMREC. No major mining finance institution would lend against a G3/G4 resource. No EPC contractor would commit to a project schedule based on it. The 5.9 million tonne figure is a geological occurrence estimate. Treating it as a strategic offset to Chinese processing dominance within this decade is not optimism. It is a classification error dressed as policy.

THE ENGINEERING BARRIER — WHY CLAYSTONE LITHIUM IS UNIQUELY DIFFICULT

The world's two proven lithium pathways: (1) Brine processing (Chile Atacama, Argentina Lithium Triangle): 1,000–3,000 ppm brines, solar evaporation 12–18 months, CAPEX $3,000–5,000/tonne LCE. (2) Spodumene hard rock (Australia Pilbara): $LiAlSi_2O_6$ ore, flotation, sulphuric acid roast, CAPEX $6,000–10,000/tonne LCE. India's J&K deposit is NEITHER. It is a sedimentary claystone deposit — lithium adsorbed onto illite and smectite clay minerals at 1,500–5,900 ppm. Claystone extraction at industrial scale has not been commercially demonstrated anywhere in the world as of 2026. Thacker Pass (Nevada, a claystone deposit) — closest global comparison — began permitting in 2016, received its Record of Decision in 2023, is only now entering first production after a decade and $1B+ in engineering investment. Additional J&K complications: Seismic Zone IV, altitude 2,500–3,500m, no industrial infrastructure within hundreds of km, Chinese hydromet technology partners geopolitically excluded. Realistic minimum timeline: 13–20 years to first battery-grade output.

THE EXPORT-REIMPORT VALUE TRAP

Even if J&K ore became available — which the timeline above demonstrates it cannot within this decade — India faces the value destruction trap without domestic refining. Battery-grade LiOH-H2O specification: minimum 56.5% purity, metallic impurities below 20–100 ppm. Achieving this requires rotary kiln calcination, membrane electrolysis or causticisation, and multi-stage recrystallisation — a capital-intensive plant taking 4–7 years to commission from greenfield. Without this: spodumene concentrates export at $800–1,200/tonne. LiOH-H2O reimports at $14,000–22,000/tonne. The value addition is 10–20x — and every dollar of that value addition India exports to a foreign refiner is a dollar of strategic leverage surrendered for extended industrial cycles.

06 Leontief Contagion Mechanics — Operationalised

Supply disruptions do not stay contained. They propagate through industrial linkages in three successive waves. The Leontief framework makes visible exactly which sectors are hit, in what sequence, and at what magnitude.

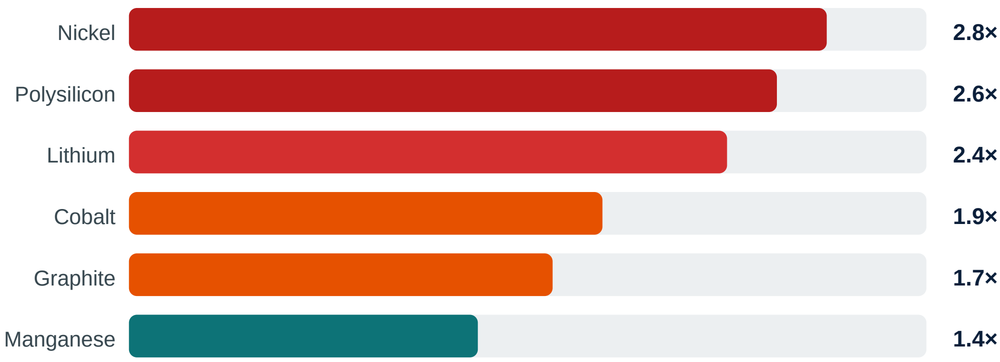
THREE-WAVE PROPAGATION MODEL

Wave 1 — Direct effects (Days 0–30): Sectors immediately deprived of the input. Cathode manufacturers cannot source NiSO4. Solar installers cannot source polysilicon. The physical supply stops. Wave 2 — Indirect effects (Days 30–90): Sectors depending on Wave 1 sectors. EV manufacturers cannot source battery packs, assembly lines halt, steel demand for chassis drops, logistics sector loses freight volume. Each transmission is multiplied by the Leontief output multiplier. Wave 3 — Induced effects (Months 3–12): Economy-wide CapEx effects. Gigafactory projects are suspended. Green infrastructure financing dries up. India's energy transition timeline slips 3–7 years. Stranded CapEx triggers broader investment confidence erosion.

Input-Output Dependency Matrix — India Electrification Supply Chain

Cell values = import dependency coefficient (0–1). >0.70 = critical chokepoint. Tier B — derived from IEA I-O coefficients.

Material	EV Batteries	Solar PV	Stainless Steel	Aerospace	Grid Storage
Nickel (Indonesia/HPAL)	**0.85**	0.05	**0.72**	0.68	0.45
Lithium (via China refining)	**0.92**	0.08	0.03	0.05	**0.88**
Polysilicon (China)	0.04	**0.95**	0.01	0.02	0.12
Cobalt (DRC/China)	**0.78**	0.02	0.08	0.52	0.65
Graphite (China)	**0.88**	0.03	0.02	0.04	**0.82**
Manganese (S.Africa/China)	0.42	0.02	0.38	0.15	0.35



Leontief output multipliers — $1 supply disruption generates $X downstream industrial output impact (Tier B)

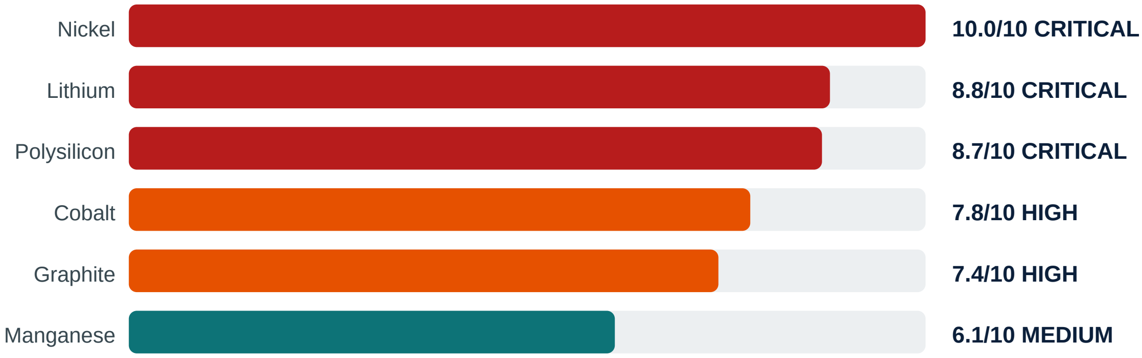
07 CIIP Battery Index v4.1 — Scoring Matrix

CIIP scores are expert-calibrated on a Leontief I-O theoretical basis. They are not regression-derived statistical outputs. Score = (EV Demand x 0.35) + (Cross-Industry x 0.40) + (Geopolitical Risk x 0.25) x (10/3).

METHODOLOGY PRECISION — PCA HAS NOT BEEN EXECUTED ON THIS DATASET

The correct characterisation: expert-calibrated with a PCA-informed weight rationale. The weight structure (EV 0.35 / Cross-Industry 0.40 / Geopolitical 0.25) is designed for compatibility with eventual PCA validation. PCA on supply disruption time-series data would likely confirm EV Demand as primary loading, Cross-Industry Competition as the dominant Leontief contagion mechanism, and Geopolitical Risk as the HHI-concentration dimension. The 90-day empirical upgrade pathway is specified in Section 12.

Material	Primary Sectors	EV	Cross	Geo	CIIP Score	Tier
Nickel	EV batteries + stainless steel + aerospace turbine blades (Inconel)	**3.0**	**3.0**	**3.0**	**10.0**	**CRITICAL**
Lithium	EV batteries + grid storage + consumer electronics	**3.0**	**2.5**	**2.8**	**8.8**	**CRITICAL**
Polysilicon	Solar PV + semiconductors + optoelectronics	**2.8**	**2.8**	**3.0**	**8.7**	**CRITICAL**
Cobalt	EV batteries (NMC) + aerospace alloys + defence electronics	**2.5**	**2.0**	**3.0**	**7.8**	**HIGH**
Graphite	Battery anode + steelmaking + nuclear moderator	**2.5**	**1.5**	**3.0**	**7.4**	**HIGH**
Manganese	NMC batteries + steel alloys + fertiliser production	**2.0**	**2.0**	**2.0**	**6.1**	**MEDIUM**



08 Chemistry Longevity Clause

This Is Not a Lithium Calculator. It Is an Inflexible Infrastructure Dependency Matrix.

THE CHEMISTRY LONGEVITY CLAUSE — FORMAL STATEMENT

The CIIP Battery framework is not a nickel or lithium calculator. It is an Inflexible Infrastructure Dependency Matrix that models midstream chemical processing chokepoints required to manufacture any high-energy-density electrochemical cell — regardless of specific anode-cathode chemistry. Whether NMC, LFP, or layered oxide sodium-ion, the manufacturing pathway requires: sulphated metal precursors, co-precipitation reactors with $NaOH/NH_4OH$ chemistry, controlled atmosphere calcination, dry room cell assembly, and formation cycling infrastructure. China controls the majority of global capacity at every one of these process steps. A chemistry transition does not exit the infrastructure dependency. It relocates the specific minerals at risk while preserving the structural chokepoint architecture.

Chemistry	Precursor Inputs	Synthesis Route	Sulphation China %	Cell Equip China %	Dependency Status
NMC cathode	$NiSO_4 + CoSO_4 + MnSO_4$	$NaOH/NH_4OH$ CSTR	82%	87%	FULL DEPENDENCY
LFP cathode	$Li_2CO_3 + FePO_4$	NaOH synthesis route	0%	85%	PARTIAL — Ni/Co eliminated
Na-ion (layered oxide)	Mn/Fe/Cu sulphates	$NaOH/NH_4OH$ CSTR	68%	85%	MIGRATED — same bottleneck
Na-ion (PBA only)	$FeSO_4$ + Na ferricyanide	Simple co-precipitation	0%	70%	REDUCED — bounded to 160 Wh/kg

LFP AND SODIUM-ION — PRECISELY BOUNDED

LFP eliminates $NiSO_4$ and $CoSO_4$ dependency entirely. It retains lithium carbonate dependency (74% Chinese refining), Chinese-dominated cell assembly equipment, and Chinese cell manufacturing IP. LFP reduces but does not eliminate infrastructure dependency. Sodium-ion (layered oxide P2/O3-type — the energy-density-competitive chemistry) uses pCAM synthesis with NaOH precipitation and NH_4OH morphology control — structurally identical to NMC. The same Chinese-dominated sulphation and CSTR infrastructure applies. Sodium-ion migrates but does not escape the dependency. The bounded exception: PBA-based sodium-ion (CATL Gen 1) uses simpler synthesis without CSTR morphology control. Achieves ~160 Wh/kg — restricting it to low-speed vehicles and stationary storage. For everything above 160 Wh/kg, the clause holds materially.

09 India Industrial Execution Feasibility

The CIIP framework identifies what India needs to build. This section models the operational constraints that determine whether and when India can build it. These are engineering and economic realities, not pessimistic projections.

Area	Key Constraints	Primary Barriers
Refining	Energy: 30–40 MWh/t LiOH. India $0.09/kWh vs China $0.03/kWh — 3× cost gap. CAPEX $2–3B for 20,000 MTPA. Environmental clearance 18–24 months.	Energy cost 3× gap · Permitting 18–24mo
Manufacturing	Gigafactory 10 GWh: $600–900M. Dry room <1% RH dewpoint — monsoon climate burden. Grid reliability <0.5% downtime vs India's 2–3% actual.	Dry room cost · Grid gap · HF infra absent
Precursors	$NiSO_4$, $CoSO_4$, $MnSO_4$ — zero domestic production. pCAM reactor is domestic. Its inputs are entirely imported. Separator membranes 95%+ import.	Metal sulphates zero domestic
Workforce	~200 battery chemical engineers/year produced. Need 2,000+ for 50 GWh scale. Electrochemistry PhDs: <50/year in relevant disciplines.	Chemical engineers 10× gap
Financing	PLI battery: 50 GWh target → 1.4 GWh delivered FY2025 (97% underdelivery). Indian gigafactory IRR 10–14% vs IRA-supported USA 16–22%.	PLI 97% underdelivery · IRR gap
J&K; Specific	G3/G4 UNFC. Seismic Zone IV. No industrial infrastructure within 200km. Claystone technology unproven globally. Chinese hydromet partners excluded.	G3/G4 unbankable · Claystone unproven

India Execution Timeline — Best / Base / Delayed Case

Milestone	Best Case	Base Case	Delayed Case
J&K; auction awarded	2027	2028	2030+
Adani Mundra polysilicon	2027	2029	2031
First 5 GWh cell output	2028	2029	2031
25 GWh domestic cell capacity	2030	2031	2034
Pilot Li refinery (non-J&K; feedstock)	2030	2032	2035
60 GWh cell capacity	2031	2033	2036
Battery-grade LiOH commercial	2032	2034	2037
J&K; first commercial ore	2038+	2040+	2042+

10 Strategic Mitigation Framework

A world-class strategic paper does not only diagnose the sickness. It prescribes the cure with operational precision. Four mitigation pathways, each targeting a specific layer of the dependency architecture.

Mitigation	Description	Specific Action
Australia CMIN Off-Take	Target BHP Nickel West MHP (Variant A) + IGO/Tianqi TLEA LiOH-H2O. Only large-scale Variant A nickel source with genuinely Western UBO. Formalise through India-Australia CMIN government-to-government framework with 5–10 year corporate off-take agreements.	Negotiate direct off-take with BHP Nickel West and TLEA. Target 2026–2027 contract execution.
Belgium/Umicore Provenance Clauses	India already sources 65%+ NiSO4 from Belgium — existing practice. Upgrade required: contractual feedstock provenance disclosure clauses requiring minimum X% of NiSO4 feedstock from non-Russian, non-Chinese-beneficially-owned sources.	Insert UBO provenance clause into all NiSO4 renewal negotiations. Engage Umicore on supply chain traceability programme.
LFP/Na-ion Chemistry Adjustment	For commercial fleets, grid storage, and 2/3-wheelers: LFP eliminates NiSO4 and CoSO4 dependency entirely. Na-ion (layered oxide) migrates to Mn/Fe precursors — lower Chinese concentration risk. This is a Layer 2 bypass for appropriate applications.	Fleet operators: mandate LFP or Na-ion in procurement specifications for sub-160 Wh/kg applications by 2027.
Hydromet Recycling Infrastructure	Invest in black mass hydrometallurgical processing co-located with gigafactory clusters (Gujarat, Pune). Recovers NiSO4, CoSO4, MnSO4, and Li2CO3 in battery-grade purity from spent cells — a secondary domestic supply loop bypassing the import dependency.	Corporate VC into black mass processing startups. Push Battery Waste Management Rules 2022 enforcement. Target 2 commercial hydromet plants by 2030.

WHAT NONE OF THESE MITIGATIONS FULLY RESOLVES

Every mitigation reduces exposure at one layer while others remain vulnerable. Australia off-take removes Variant A nickel UBO Risk — it does not address CoSO4 (Chinese-processed DRC cobalt), MnSO4 (Chinese sulphation), or graphite. Belgium NiSO4 addresses Variant B nickel — feedstock Russian/African UBO exposure requires contractual clauses not currently standard in Indian procurement. LFP chemistry switch eliminates nickel/cobalt — retains lithium carbonate (74% Chinese refining) and Chinese cell manufacturing equipment. Recycling creates a secondary supply loop — cannot materialise at scale before 2032–2035. The only complete solution is domestic midstream chemical processing infrastructure: metal sulphate production, lithium refining, and cell manufacturing equipment localisation. Everything else is mitigation, not resolution.

11 Scenario Architecture — Four Pathways

All 2030 projections are Tier C — strategic scenario assumptions, not econometric forecasts. The $19B–$53B range bounds outcome space. No single projection should be cited without its scenario label and assumption set.

Metric	S1 — China Dominance	S2 — Moderate	S3 — Aggressive	S4 — Na-ion+Recycling
Battery import FY2030	**$23B**	**$17B**	**$11B**	**$8B**
Solar equipment FY2030	**$30B**	**$21B**	**$13B**	**$11B**
Combined clean energy	**$53B**	**$38B**	**$24B**	**$19B**
India domestic cell GWh	**5 GWh**	**25 GWh**	**60 GWh**	**40 GWh**
India Li refining	**Near zero**	**Partial**	**Significant**	**Moderate**
China battery share (India)	**80%+**	**55–65%**	**35–45%**	**30–40%**
Nickel CIIP score (adjusted)	**10.0**	**8.8**	**7.5**	**5.5**

Scenario assumption note: S1 assumes no domestic refining, PLI underdelivery, China controls 80%+ midstream. S2 assumes partial PLI success, J&K; auction (non-J&K; feedstock refinery). S3 assumes Adani Mundra + J&K; operational, AU/Chile partnerships. S4 assumes Na-ion commercialisation, 20%+ recycling recovery, LFP dominant. Currency: USD constant 2025. Tier C — directional estimates only.

12 Empirical Validation Roadmap — 30 / 90 / 365 Days

CIIP scores are currently expert-calibrated. This roadmap specifies the data tracks, analytical models, and execution timeline required to achieve statistical validation.

Horizon	Deliverable	Tools/Sources	CIIP Upgrade
30 Days	USGS HHI concentration index 2015–2025 for all 6 materials	USGS Mineral Commodity Summaries	Replaces qualitative geo score
30 Days	India import data by material and source country	UN Comtrade API (free tier)	India-specific exposure layer
30 Days	LME nickel/cobalt price series 10-year monthly	Quandl / Yahoo Finance	CV-weighted volatility dimension
90 Days	PCA on assembled panel — derive empirical dimension weights	Python sklearn PCA	Replaces 0.35/0.40/0.25 expert weights
90 Days	Backtest CIIP scores against 15 documented disruption events	Export control event database	Validates scoring methodology
90 Days	UBO-adjusted concentration index (ownership vs origin)	Corporate registry + Bloomberg	Corrects country-of-origin bias
12 Months	Directed graph using NetworkX — betweenness centrality	Python NetworkX	Identifies top-5 chokepoint nodes
12 Months	Monte Carlo simulation — confidence intervals for all scores	Python scipy / numpy	Quantifies score uncertainty
12 Months	Extend to AI infrastructure, semiconductors, fertiliser minerals	IEA, USGS, UN Comtrade	India 2030 ORM — CIIP v5.0

13 Country Dependency — Origin Country vs UBO Country

Countries selected by EV growth trajectory and strategic exposure. Data uses origin country per trade statistics. UBO-adjusted exposure shows materially higher Chinese structural exposure than origin-country data suggests.

Country	EV Share	Growth	Risk	Key Dependencies (Origin + UBO Notes)
India	4% (2025)	+45% YoY	CRITICAL	Li compounds 68% China · Batteries 85% China · NiSO4 65%+ Belgium (feedstock: Russian/African) · Ni hydroxide 64.7% Australia (UBO: Western) · CoSO4 ~80% Chinese · Li refining ZERO
China	52% (2025)	+18% YoY	MEDIUM	Ni ore ~60% Indonesia (UBO: Chinese) · Li ore ~55% Australia · Co ore 70%+ DRC (UBO: Chinese) · Graphite 85% domestic · LFP cathode 98% domestic
USA	10% (2025)	-4% YoY	HIGH	Cobalt 75% DRC via China · Graphite ~70% China · Battery cells ~65% Asia · IRA re-shoring early stage
Europe	27% (2025)	+33% YoY	HIGH	Graphite ~80% China · Cobalt ~80% via China · Nickel ~60% Indo/Russia · CRM Act early implementation
Indonesia	14% (2025)	+100%+ YoY	HIGH	Ni ore domestic (67%+ global output) BUT HPAL plants UBO: Tsingshan/Huayou (Chinese) · Batteries 90% China JV
Vietnam	40% (2025)	from ~0% 2020	HIGH	Batteries 85% China · Polysilicon ~90% China origin · VinFast BOM ~60% Chinese · Re-labelling not diversification

14 Contagion Watch Signals — Weekly Monitoring

Five signals provide structural early warning of supply chain stress based on the Leontief propagation mechanics in Section 06. The CIIP framework is a live operational intelligence system.

CRITICAL	**Indonesia Nickel Quota + Tsingshan/Huayou Allocation Decisions** 67%+ of global mine output. The operationally relevant signal is not Jakarta's government quota — it is Tsingshan and Huayou's monthly MHP allocation decisions. These are corporate decisions made in Wenzhou, not government policies made in Jakarta. They do not appear in export control notifications. *Monitor: ESDM ministry quotas · ANTAM/Vale production · Tsingshan/Huayou quarterly output · LME Ni forward curve vs spot spread*
CRITICAL	**China MOFCOM/GACC Export Control Expansion — Next Frontier: Equipment** Oct 2024 controls: graphite anode, cathode precursors, LFP technology. Next escalation frontier: battery formation and ageing equipment, electrode coating machinery. Layer 3 controls cannot be mitigated by stockpiling materials. *Monitor: MOFCOM announcements weekly · GACC export licence approval rate · Chinese battery equipment industry association statements*
HIGH	**India J&K; Programme — Resource Delineation Milestones** The signal to watch is NOT auction award dates. The signal is whether GSI publishes infill drilling results that upgrade J&K; from G3/G4 to G1/G2 — the minimum for a bankable feasibility study. Without this upgrade, no serious capital will commit. *Monitor: GSI Annual General Report · MECL drilling programme updates · Ministry of Mines J&K; resource delineation budget*
HIGH	**NMC vs LFP Chemistry Split in Indian EV Registrations** India SUV/MPV segment (80% of EV sales 2025) is NMC-favourable. This split determines whether nickel CIIP holds at 10.0 or falls below 8.0. The single most important variable for nickel score trajectory to 2030. *Monitor: SIAM monthly registrations by segment · FAME-III battery chemistry disclosures · OEM powertrain specification sheets*
MEDIUM	**Adani Mundra Polysilicon — Electricity Cost and Commissioning Progress** The strategic variable is not whether Adani commissions the plant — it is whether the plant can operate competitively. Siemens process requires 30–40 kWh/kg. India industrial electricity at \$0.09–0.11/kWh creates a structural cost disadvantage vs China at \$0.03–0.05/kWh. *Monitor: Adani Green quarterly CAPEX · Gujarat RERC tariff orders · MNRE polysilicon capacity data*

REF References & Data Sources — All 22 Sources

All references carry a data confidence tier. Tier A = institutionally verified primary data. Tier B = derived analytical estimate. Tier C = strategic scenario assumption.

[R01] Tier A IEA Global EV Outlook 2025 — Electric Vehicle Batteries
https://www.iea.org/reports/global-ev-outlook-2025
Key data: China 80% battery cell production; lithium demand +30% 2024

[R02] Tier A IEA Global EV Outlook 2026 — Trends in Electric Cars
https://www.iea.org/reports/global-ev-outlook-2026
Key data: China 13M EV sales 2025; 52% market share; India SUV/MPV 80% of EV sales

[R03] Tier A IEA Critical Minerals Outlook 2025
https://www.iea.org/reports/global-critical-minerals-outlook-2025
Key data: China refining dominance: Li 74%, Co 80%, graphite 85%

[R04] Tier A USGS Mineral Commodity Summaries 2025 — Nickel
https://pubs.usgs.gov/periodicals/mcs2025/mcs2025-nickel.pdf
Key data: Indonesia 67%+ global output; Australia -26%; global 3.7Mt 2024; HHI basis

[R05] Tier A IndexBox — India Battery Manufacturing Import Dependency (2026)
https://www.indexbox.io/blog/indias-battery-storage-push-import-dependency-persists
Key data: India 18,200t lithium imports; 68% China; S1 2030 projection $23B

[R06] Tier A IEEFA — Securing India's Battery Supply Chain (2026)
https://ieefa.org/resources/securing-indias-battery-supply-chain-more-critical-ever
Key data: India Li refining zero; cobalt 2,060t/yr; J&K; G3/G4 classification

[R07] Tier A IEEFA/ICRIER — India Battery-Grade Nickel Trade Dynamics (2026)
https://ieefa.org
Key data: Variant A: Australia 64.7% of Ni oxide/hydroxide. Variant B: Belgium 65%+ of NiSO4.

[R08] Tier A ORF — India's EV Sector and the China Challenge (2025)
https://www.orfonline.org/expert-speak/india-s-ev-sector-and-the-china-challenge
Key data: India imports 90%+ Li/Co/Ni requirements; China 85% battery import share

[R09] Tier A Down to Earth — India Solar Surge: China Polysilicon Dominance
https://www.downtoearth.org.in/energy
Key data: China 91% polysilicon, 97% wafers; India $7B solar imports FY2024

[R10] Tier B Deccan Herald — India Solar Imports $30B by 2030
https://www.deccanherald.com
Key data: $30B solar projection — S1 scenario assumption only (Tier C context)

[R11] Tier A IEEFA — War Opens Renewable Energy Pathway (March 2026)
https://ieefa.org/resources/war-opens-renewable-energy-pathway
Key data: India oil import $116.4B FY2025; crude 229Mt; solar 143GW Feb 2026

[R12] Tier A ICCT — Global EV Market Monitor 2025
https://theicct.org/publication/global-ev-market-monitor-for-ldvs-2025/
Key data: Global 20M+ EVs 2025; Vietnam 40% share; India 4% share

[R13] Tier A ING Think — Nickel Still Capped by Surplus (December 2025)
https://think.ing.com/articles/nickel-still-capped-by-surplus/
Key data: NMC China share 18% H1 2025; LFP dominance structurally capping nickel demand

[R14]	Tier A	Investing News — Top Nickel Producing Countries (2025) *https://investingnews.com/nickel-investing/top-nickel-producing-countries/* *Key data: Indonesia, Philippines, Russia top producers — not Australia in production ranking*
[R15]	Tier A	Statranker — Top 10 Nickel-Producing Countries 2025 *https://statranker.org* *Key data: Indonesia 2.6Mt 2025; 67%+ global share; global production 3.9Mt*
[R16]	Tier A	ScienceDirect — Nickel-Based Superalloys for Gas Turbines (2023) *https://www.sciencedirect.com/science/article/abs/pii/S0925838823024313* *Key data: Single-crystal Inconel exclusively for turbine blades; no substitute above 1,000C*
[R17]	Tier A	Academia.edu — Nickel Base Superalloys for Aero Engine Turbine Blades *https://www.academia.edu/26183228* *Key data: Turbine inlet temperature determined exclusively by Ni-base superalloy capability*
[R18]	Tier A	Takshashila — India's Options Amid Chinese Export Curbs (2025) *https://takshashila.org.in* *Key data: China Oct 2024 controls: graphite, LFP cathode, cell tech; India $2.2B cell imports*
[R19]	Tier A	Stainless Steel World — World Nickel Market in 2025 *https://stainless-steel-world.net* *Key data: 70% nickel in stainless steel; NMC811 ~40% 2024; battery demand growing*
[R20]	Tier A	LKY School — China Factor in India's Battery Ambitions (2025) *https://lkyspp.nus.edu.sg* *Key data: Amara Raja-Gotion; Exide-SVOLT; India equipment imports $1.5B 2022*
[R21]	Tier A	Investing News — Top 9 Countries by Nickel Reserves *https://investingnews.com/nickel-investing/nickel-reserves-by-country/* *Key data: Indonesia, Australia, Brazil top reserves — reserves vs production distinction*
[R22]	Tier A	Lithium Americas — Thacker Pass Technical Reports (2023–2025) *https://www.lithiumamericas.com/thacker-pass/* *Key data: Claystone Li benchmark: 9+ years permitting to production; $1B+ engineering*

METHODOLOGICAL DISCLAIMER: CIIP scores are expert-calibrated analytical judgements grounded in Leontief Input-Output economics. They are not regression-derived statistical outputs. 'PCA-informed' means the weight structure is designed for compatibility with future PCA validation — PCA has not been executed on this dataset as of v4.1. All 2030 projections are Tier C scenario assumptions — directional estimates bounded by named assumptions, not econometric forecasts. UBO risk assessments are based on publicly available corporate ownership information and reflect analytical judgement. J&K; lithium occurrence is classified G3/G4 under UNFC — it is a geological notation, not a bankable reserve. Developer: Aditya V Sivaram Poduri · India Supply Chain Signals · indiasupplychainsignals.substack.com